

National Institute of Technology Rourkela
Department of Computer Science and Engineering
End-Term Examination (May 2024)
Computer Vision (CS3303), B.Tech. 6th Semester

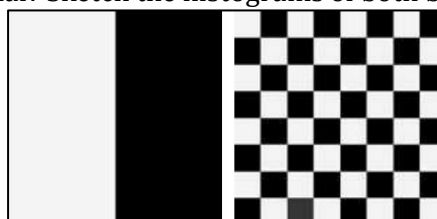


Time: 180 Minutes

Max. Marks: 50

- Instructions:** 1. In case of any doubt, write your assumptions, write it clearly and continue.
2. There are 08 questions printed on the paper. Attempt all the questions.
3. Please write the answers of one question (all subparts) in sequence only.

- Q1.** [A] Write down two prominent references you have followed to prepare two specific topics covered in the lectures. [2]
[B] What is the requirement of sampling and quantisation? Explain how you will decide the sampling rate and quantisation bits. [2]
[C] Explain how a human vision system works and compare its components with the respective components of a digital camera. [4]
- Q2.** [A] Explain with an example why sharpening filters based on the second derivative will produce more emphasised edges compared to the filter based on the first derivative. [2]
[B] What if the histogram equalisation is applied twice on an image? Explain. [2]
[C] The images below, left and right, are quite different, but their histograms are the same. Suppose that each image is blurred with an averaging mask. Would the histograms of the blurred images still be equal? Sketch the histograms of both blurred images. [2]



Left

Right

- Q3.** [A] What activation functions can be used in the CNN models? Explain their advantages and disadvantages. [2]
[B] Transfer learning (TL) is a research problem in machine learning (ML) that focuses on storing knowledge gained while solving one problem and applying it to a different but related problem. Explain different cases/approaches for transfer learning. [4]
- Q4.** [A] How does the Canny Edge detector use the direction of gradient information? Explain the Canny edge detection pipeline. [4]
[B] Explain the Harris corner detection method with all the steps. [2]
- Q5.** [A] Explain a straight-line detection technique based on the Hough Transform. Briefly discuss its advantages and disadvantages. [4]
[B] Why do we prefer p-theta (polar) parameter space over m - c parameter space for the Hough transform? [2]
- Q6.** [A] SIFT is a feature descriptor used as a key point descriptor of an image in the scale-space domain. Present an analysis of various steps to match the objects in different images using SIFT. [4]
[B] Describe how the RANSAC algorithm could be used for inlier correspondences in order to address the outlier issues. [2]

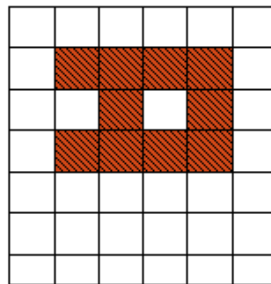
- Q7.** A binary array that represents a portion of a black-and-white image is given below. Perform the operations listed below on this piece of image. Assume that all of the pixels that surround this segment contain black (0) background. The structuring element is shown below where the origin is shown as circled element. **[6]**
(4x1+2)

0	0	0	0	0	0	0
0	0	1	1	0	0	0
0	0	0	1	0	0	0
0	0	0	1	1	0	0
0	0	1	1	1	1	0
0	0	1	1	1	0	0
0	1	0	1	0	1	0
0	0	0	0	0	0	0

1	Ⓐ
0	1

Structuring
Element

- [A] Dilation with the structuring element.
 [B] Erosion with the structuring element.
 [C] Opening with the structuring element.
 [D] Closing with the structuring element.
 [E] Hit and miss transform with the structuring element.
- Q8.** Explain the steps, with the output of the respective step, to obtain the convex hull for the figure shown below. (The filled pixels are representing the pixel value 1) **[6]**



End of Paper